

Series Sheath Capacitance of a Plasma-Immersed Dipole in Three-Dimensional Fluid FDTD: Volumetric Jacket Versus Analytic Coaxial Loading

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Key points

- A conductor-coated vacuum jacket in 3D fluid FDTD loads the feed as series sheath capacitance, not a shunt capacitor.
- Dense CW at all $S_d = 0, 2, 4, 6, 8, 10$ shows monotonic capacitive loading; $\text{Im}\{Z\}$ crossings fall with sheath width.
- Band-averaged $C_{\text{eff}}(S_d)$ tracks coaxial C_{sh} in trend, with a scale factor near 0.3.

Abstract

We quantify how a prescribed volumetric vacuum sheath of width S_d cells loads the feed impedance of a thin-wire dipole in a three-dimensional plasma-fluid finite-difference time-domain (PF-FDTD) solver. The sheath is not a lumped element placed across the feed. It is a conductor-attached density hole: ambient density is depleted by 10^{-6} in a stair-cased shell of thickness $S_d\Delta x$ about every PEC cell, so plasma conduction current cannot reach the metal and feed current must close by displacement current across the radial gap. That geometry is the FDTD analogue of a series sheath capacitor in series with the plasma-loaded antenna, not a shunt feed capacitance. Continuous-wave (CW) phasors at plasma frequency $f_p = 2$ MHz are reported for the full thickness set $S_d \in \{0, 2, 4, 6, 8, 10\}$ in 1.50–2.30 MHz. For every $S_d \geq 2$, $\text{Im}\{Z\} < 0$ throughout that band and $\text{Im}\{Z\}$ at fixed frequency becomes steadily more negative as S_d increases. A low-frequency extension at $S_d = 0, 2, 10$ (0.50–1.50 MHz) was run because those three thicknesses already showed that any $\text{Im}\{Z\} + \rightarrow -$ crossing for $S_d \geq 2$ lies below 1.50 MHz; $S_d = 4, 6, 8$ were not omitted from the physics, only from the extra low-frequency grid. The first $+ \rightarrow -$ marker is $f_{\text{res}}(0) = 1.846$ MHz, $f_{\text{res}}(2) = 0.655$ MHz, $f_{\text{res}}(4, 6, 8) < 1.50$ MHz, and $f_{\text{res}}(10) \leq 0.50$ MHz. Defining $C_{\text{eff}} = -1/(\omega \text{Im}\{Z\})$ on capacitive branches and averaging 0.8–1.2 MHz gives $C_{\text{eff}}(2) \approx 7.4$ pF and $C_{\text{eff}}(10) \approx 4.3$ pF. Dense-band averages at 1.50–1.70 MHz for $S_d = 4, 6, 8, 10$ continue the same decrease. Both sets track coaxial $C_{\text{sh}} = 2\pi\epsilon_0 L / \ln(1 + S_d\Delta x/r_{\text{eff}})$ in trend, with $C_{\text{eff}}/C_{\text{sh}} \approx 0.3$ in the quasi-static window. The work is a static prescribed-jacket result, not a claim of Tu/Song dynamic sheath fidelity.

Plain Language Summary

Radio antennas in space plasmas sit inside a thin empty sheath next to the metal. We simulated that empty layer as a jacket of controlled thickness around a dipole in a 3-D plasma wave code. The jacket wraps the wire, so current has to jump the empty gap to reach the plasma; that is a series capacitor, not a capacitor across the antenna terminals. Making the jacket thicker made every thickness we simulated look more capacitive and moved the resonance to lower frequency. The effective capacitance extracted from the simulations decreases with jacket thickness in the same way as a simple coaxial capacitor formula, up to a nearly constant scale factor.

1 Introduction

A conducting antenna immersed in plasma is surrounded by a depleted sheath that can dominate low-frequency impedance. Classical short-dipole theory in a cold magnetoplasma (Balmain, 1964, 1969) provides the homogeneous-medium baseline; later Balmain-era and contemporary treatments add ion sheaths, insulating jackets, and sheath-wave physics as series gap loading between metal and exterior plasma. Modern plasma impedance probe (PIP) retrievals often write an explicit coaxial sheath capacitor in series with a Balmain-like plasma term (Liu et al., 2025; Brooks & Paliwoda, 2024), so that the phase-zero frequency lies below bulk plasma features.

High-voltage transmitting antennas raise a related but distinct problem: Song and coworkers modeled an oscillating ion-sheath radius about a bare dipole (Song et al., 2007), and Tu et al. used one-dimensional particle-in-cell (PIC) simulations to refine sheath structure and reactance (Tu et al., 2008). Those studies motivate the statement that sheath changes antenna impedance, but they do not prescribe a fixed vacuum jacket in three-dimensional fluid FDTD, nor do they require that an $\text{Im}\{Z\} = 0$ marker climb with sheath thickness.

This paper addresses a narrower, quantitative question suited to the PF-FDTD solver used here (Ward, 2006): does a *prescribed* volumetric vacuum jacket of width $S_d\Delta x$ behave as a *series* sheath capacitor whose effective $C_{\text{eff}}(S_d)$ tracks coaxial analytic expectations? Campaign diagnostics that established conductor-attached sheath coupling are summarized only as needed; the methodological path, including a pre-fix seeding bug, is documented elsewhere (Neshragh, 2026). Success is defined by C_{eff} versus coax and multi-marker reporting of $Z(f)$, not by an upward Im-zero shift with S_d .

2 Methods

2.1 PF-FDTD solver and antenna geometry

PF-FDTD advances Maxwell’s equations on a staggered Yee grid coupled to multi-species fluid continuity and momentum equations (Ward, 2006). Plasma current enters Ampere’s law; feed observables are voltage $V(t)$ and current $I(t)$ at a hard E_z source, with

$$Z(f) = \frac{V(f)}{I(f)} \tag{1}$$

from steady-state CW phasors on the last 50% of each `.vc` record.

All runs use a $70 \times 70 \times 65$ grid with $\Delta x = 0.04$ m and a z -directed thin-wire dipole from grid indices (35, 35, 20) to (35, 35, 45) (length $L = 1.00$ m) with the feed at (35, 35, 33). Unless noted, $f_p = 2$ MHz, collisions are weak, and the background magnetic field is zero.

2.2 Prescribed volumetric sheath as a series gap

The sheath is imposed as a *jacket that coats the conductor*, not as a circuit element connected across the feed terminals (Figure 1). Ambient density is initialized uniformly. After the antenna permittivity flags exist, a discrete distance field is grown from every PEC cell ($\text{ER}_x \text{ER}_y \text{ER}_z = 0$). Cells whose distance d (in grid steps) satisfies

$$0 < d \leq S_d \quad \Rightarrow \quad N_0 \rightarrow N_0 \cdot 10^{-6} \quad (2)$$

are depleted to a numerical floor ($N_{\min}/N_0 = 10^{-6}$). Plasma conduction current in that shell is negligible, so the jacket behaves as vacuum. Seeding from the plasma-on mask rather than from PEC previously placed the hole at the domain boundary and made Z independent of S_d ; all results here use the PEC-seeded implementation (Neshragh, 2026). Figure 2 shows the resulting radial step in N_0 .

Because the depleted shell surrounds the metal, feed current reaching the bulk plasma must cross the radial gap as displacement current. The corresponding circuit is

$$Z_{\text{in}}(f) \approx Z_{\text{pl}}(f) + \frac{1}{j\omega C_{\text{sh}}(S_d)}, \quad (3)$$

i.e. series sheath capacitance, not a shunt C across the terminals. Parallel-plate estimates $C \sim \epsilon_0 A / (S_d \Delta x)$ are only pedagogical magnitude guides for that series gap. Thin-wire geometry favors the logarithmic coaxial form used below.

2.3 CW campaigns and why the low-frequency grid is smaller

Two CW grids share the same geometry and f_p :

Dense band (all thicknesses) $S_d \in \{0, 2, 4, 6, 8, 10\}$, $f = 1.50$ – 2.30 MHz (100 kHz and selected half steps; 66 completed cases). This is the full S_d survey.

Low-frequency extension $S_d \in \{0, 2, 10\}$, $f = 0.50$ – 1.50 MHz (100 kHz steps). These three thicknesses were re-run below 1.50 MHz after the dense band showed that

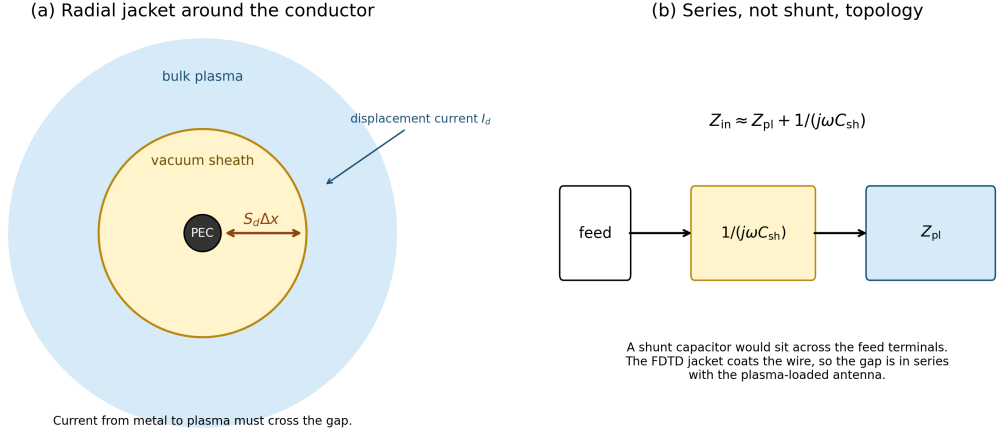


Figure 1. How the prescribed jacket mimics a series sheath. (a) Cross-section: a vacuum shell of thickness $S_d \Delta x$ coats the PEC wire; bulk plasma lies outside. (b) Equivalent topology: displacement current across the gap inserts $1/(j\omega C_{sh})$ in series with the plasma-loaded antenna Z_{pl} . A shunt capacitor would sit across the feed, which is not the geometry we impose.

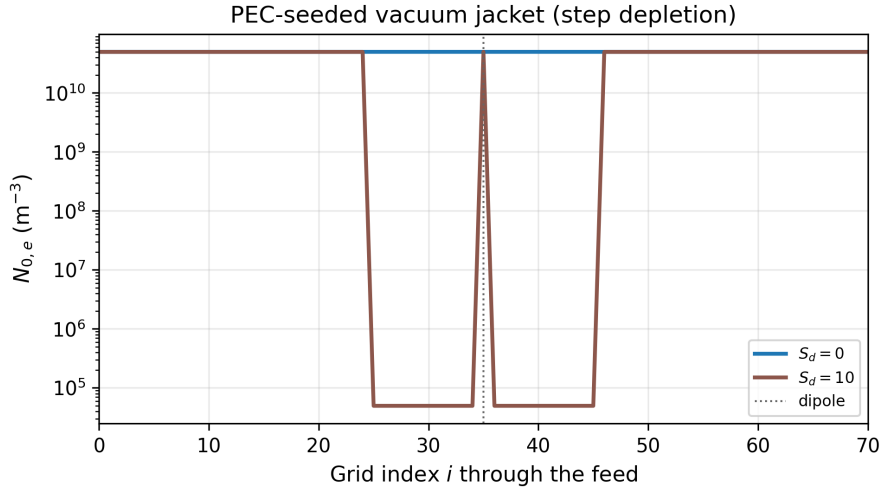


Figure 2. Schematic of the applied electron-density mask along a cut through the feed. For $S_d = 10$ the PEC-seeded step depletes N_0 by 10^{-6} in ten cells on either side of the wire; $S_d = 0$ leaves a uniform plasma.

every $S_d \geq 2$ was already capacitive there, so any $+$ $\rightarrow$ $-$ crossing had to lie at lower frequency. Intermediate widths $S_d = 4, 6, 8$ remain in the dense-band results; they were not discarded.

One dense-band sample ($S_d = 0, 1.50$ MHz) is an incomplete restart file and is omitted from all plots and averages.

Series resonance f_{res} is the first $\text{Im}\{Z\}$ positive-to-negative zero crossing by linear interpolation. We also report $|Z|$ in the dense band as a second marker and do not equate phase-zero with magnitude extrema.

2.4 Effective capacitance and coaxial reference

Where $\text{Im}\{Z\} < 0$, define

$$C_{\text{eff}}(f) = -\frac{1}{\omega \text{Im}\{Z(f)\}}, \quad \omega = 2\pi f. \quad (4)$$

This C_{eff} is an *effective* feed capacitance of the full antenna–sheath–plasma system, not a pure vacuum gap to an ideal plasma electrode. The quasi-static average uses 0.8–1.2 MHz (capacitive points only), away from the narrow inductive island of $S_d = 2$ near 0.60 MHz. That window exists only for the low-frequency set $S_d = 2, 10$. For $S_d = 4, 6, 8$ (and as a consistency check on $S_d = 10$) we also average C_{eff} over 1.50–1.70 MHz, the lowest dense-band frequencies still well below f_p . We do not use the dense-band C_{eff} for $S_d = 2$, where $\text{Im}\{Z\}$ is approaching zero and equation (4) is no longer a quasi-static capacitor.

The coaxial analytic sheath for a cylinder of length L and effective radius r_{eff} is

$$C_{\text{sh}}^{(\text{coax})} = \frac{2\pi\epsilon_0 L}{\ln(1 + S_d \Delta x / r_{\text{eff}})}. \quad (5)$$

We take $r_{\text{eff}} = 0.23\Delta x$ as a Holland-type thin-wire estimate on the Yee lattice (Holland & Simpson, 1981); sensitivity to r_{eff} is discussed in Section 4.

3 Results

3.1 Dense-band impedance at all S_d

Figure 3 shows the full thickness survey. Curves are ordered with S_d : larger jackets reduce $\text{Re}\{Z\}$ and drive $\text{Im}\{Z\}$ more negative at every frequency in 1.50–1.80 MHz. For $S_d \geq 2$, $\text{Im}\{Z\} < 0$ at all dense-band tones; only $S_d = 0$ crosses from inductive

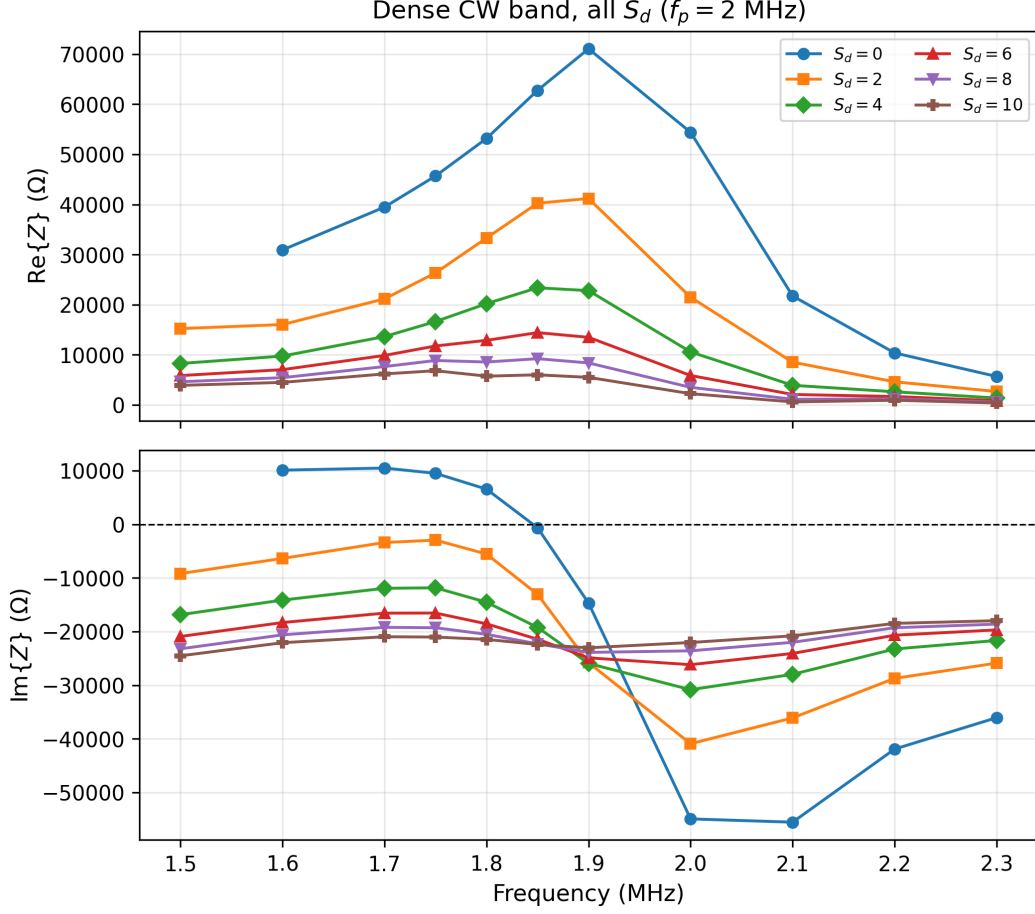


Figure 3. Dense-band CW phasor impedance for all simulated jackets $S_d = 0, 2, 4, 6, 8, 10$ ($f_p = 2$ MHz). Top: $\text{Re}\{Z\}$; bottom: $\text{Im}\{Z\}$. The $S_d = 0, 1.50$ MHz restart file is omitted. Every $S_d \geq 2$ is capacitive throughout this band.

to capacitive, at 1.846 MHz. The same ordering appears in $|Z|$ (Figure 4): the dense-band magnitude peak collapses monotonically as the jacket thickens. Intermediate widths $S_d = 4, 6, 8$ lie between $S_d = 2$ and $S_d = 10$; they are not outliers and are not omitted from the loading argument.

3.2 Low-frequency extension for $S_d = 0, 2, 10$

Figure 5 stitches the low-frequency extension onto the dense band for the three thicknesses that were re-run below 1.50 MHz. $S_d = 0$ stays inductive through 0.50–1.50 MHz. $S_d = 10$ is capacitive at every low-frequency point. $S_d = 2$ is capacitive except for a brief inductive island at 0.60 MHz, with a $+\rightarrow-$ crossing at 0.655 MHz.

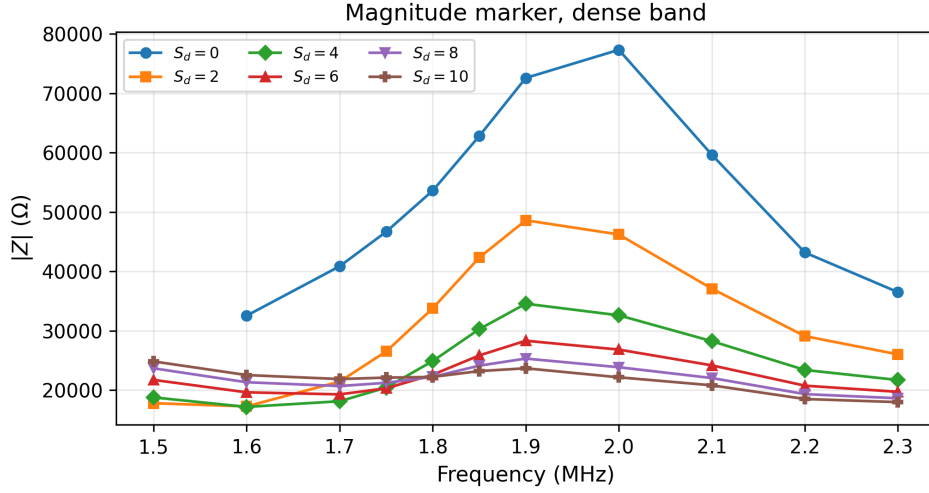


Figure 4. Dense-band $|Z|$ for the same six thicknesses. The near- f_p magnitude peak decreases monotonically with S_d and is a second marker, distinct from the $\text{Im}\{Z\} = 0$ crossing.

Table 1. Series resonance from $\text{Im}\{Z\} +$ to $-$ crossings

S_d	f_{res}	Note
0	1.846 MHz	Dense band; low-frequency band all inductive
2	0.655 MHz	Also $-$ to $+$ at 0.570 MHz (inductive island)
4, 6, 8	< 1.50 MHz	All $\text{Im}\{Z\} < 0$ in the dense band; no low-frequency run
10	≤ 0.50 MHz	All $\text{Im}\{Z\} < 0$ in 0.50–2.30 MHz

3.3 Resonance markers versus sheath width

Table 1 and Figure 6 collect $\text{Im}\{Z\} + \rightarrow -$ crossings. Widening the jacket moves f_{res} *downward*. For $S_d = 4, 6, 8$ the dense band supplies only an upper bound (< 1.50 MHz), consistent with the measured $S_d = 2$ and $S_d = 10$ crossings.

3.4 Effective capacitance versus coaxial sheath

Figure 7 shows $C_{\text{eff}}(f)$ from equation (4) on capacitive low-frequency branches. Band means are listed in Table 2. Figure 8a confirms that, at the single dense-band frequency 1.60 MHz, $\text{Im}\{Z\}$ for $S_d = 2, 4, 6, 8, 10$ is a monotonic function of thickness. Figure 8b compares C_{eff} with equation (5).

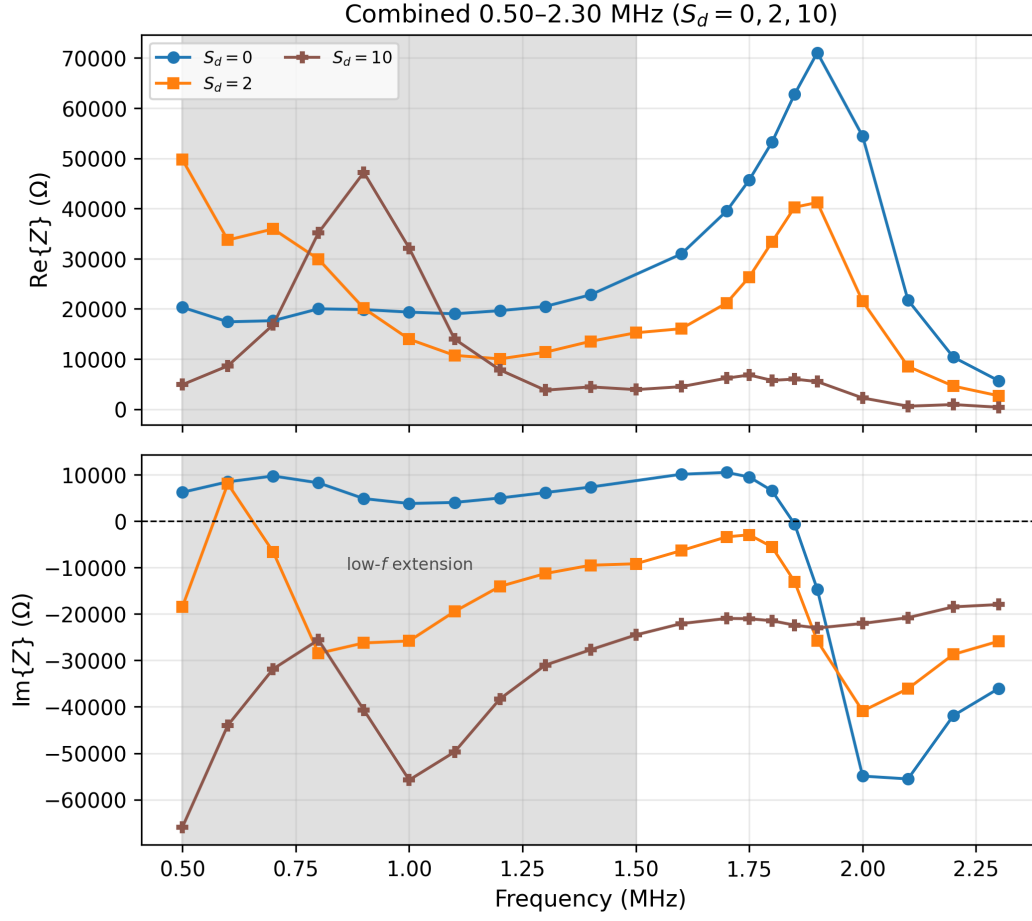


Figure 5. Combined CW impedance for $S_d = 0, 2, 10$ over 0.50–2.30 MHz. The shaded region is the low-frequency extension, run because the dense band (unshaded) already showed $S_d \geq 2$ to be capacitive.

Table 2. Band-averaged C_{eff} versus coaxial C_{sh} . Low-frequency window: 0.8–1.2 MHz. Dense window: 1.50–1.70 MHz.

S_d	t_{sh} (m)	$C_{\text{eff}}^{(\text{low})}$ (pF)	$C_{\text{eff}}^{(\text{dense})}$ (pF)	$C_{\text{sh}}^{(\text{coax})}$ (pF)	Ratio
2	0.08	7.35 ± 1.12	—	24.49	0.30
4	0.16	—	7.06 ± 0.63	19.11	0.37
6	0.24	—	5.39 ± 0.24	16.86	0.32
8	0.32	—	4.76 ± 0.13	15.55	0.31
10	0.40	4.27 ± 1.83	4.43 ± 0.08	14.66	0.29

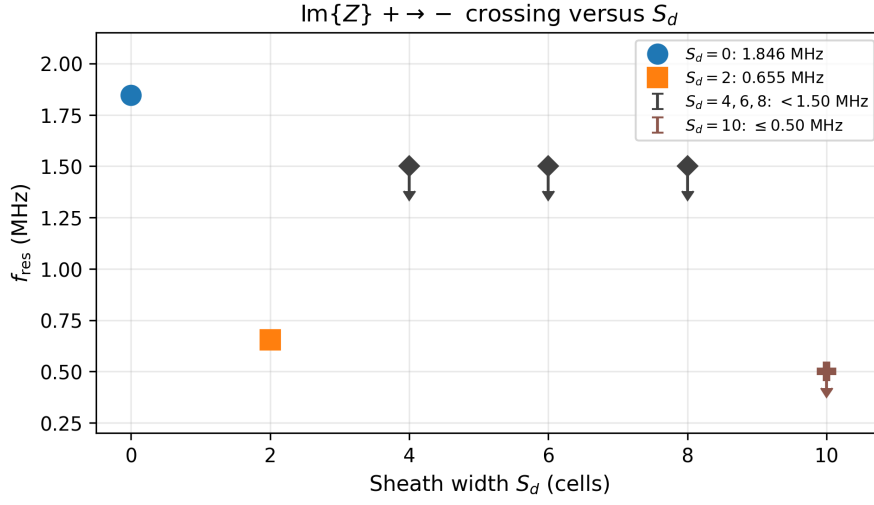


Figure 6. Resonance frequency $f_{\text{res}}(S_d)$ from $\text{Im}\{Z\}$ positive-to-negative crossings. Diamonds and the $S_d = 10$ symbol are upper limits (no crossing in the available band), not missing data.

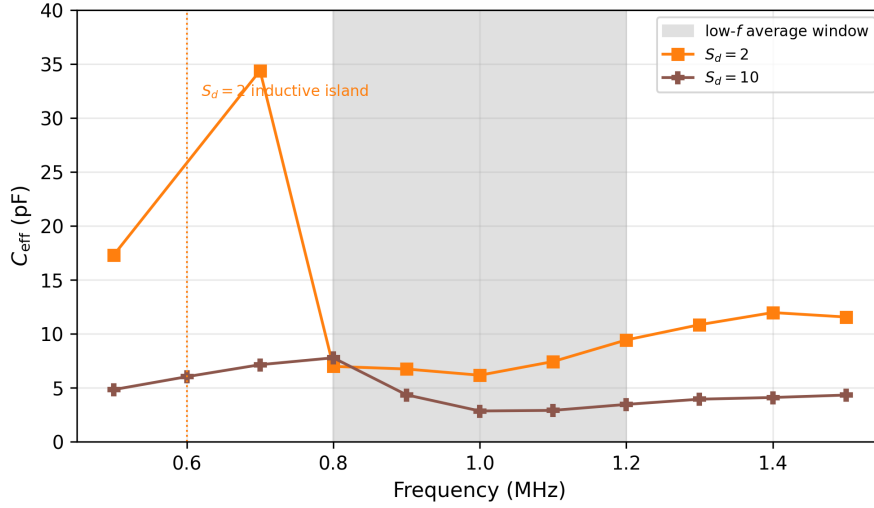


Figure 7. Effective capacitance $C_{\text{eff}}(f)$ for $S_d = 2$ and 10. The shaded band is the averaging window in Table 2. The spike near 0.70 MHz for $S_d = 2$ is the approach to the inductive island and is excluded from the mean.

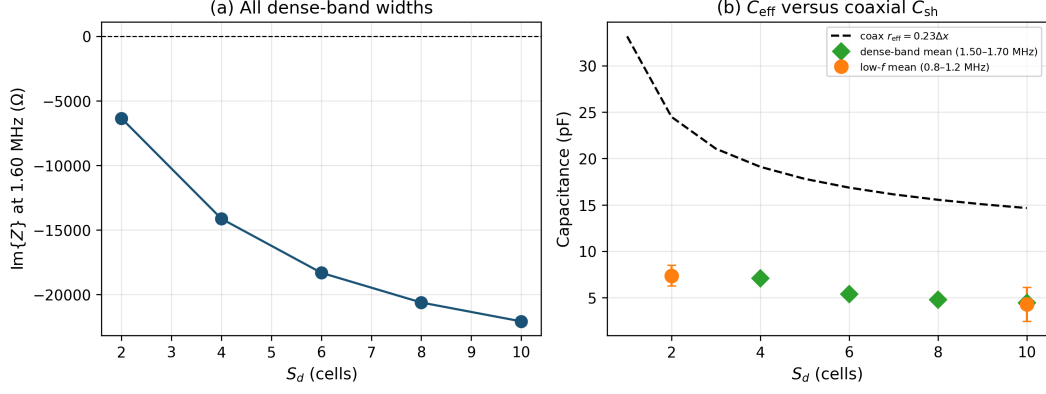


Figure 8. (a) Dense-band $\text{Im}\{Z\}$ at 1.60 MHz for every jacket with $S_d \geq 2$: thicker sheaths are more capacitive. (b) $C_{\text{eff}}(S_d)$ compared with coaxial analytic C_{sh} for $r_{\text{eff}} = 0.23\Delta x$. Orange: quasi-static low-frequency means. Green: dense-band means for widths that were not re-run below 1.50 MHz (plus $S_d = 10$ as a cross-check). Dense-band $C_{\text{eff}}(S_d = 2)$ is omitted because $\text{Im}\{Z\}$ is approaching zero in that window.

Low-frequency ratios $C_{\text{eff}}/C_{\text{sh}}$ are 0.30 ($S_d = 2$) and 0.29 ($S_d = 10$). Dense-band ratios for $S_d = 4, 6, 8, 10$ lie in 0.29–0.37. Absolute scale therefore depends on r_{eff} , stair-casing, and residual plasma reactance inside C_{eff} , but the *trend* with S_d matches series coaxial loading: thicker jackets reduce capacitance and strengthen capacitive reactance at fixed frequency.

4 Discussion

4.1 Series versus parallel topology

The depleted shell sits between the conductor and bulk plasma (Figure 1). Equation (3) follows from that geometry, not from a post-hoc circuit fit. A shunt capacitor across the feed would not thicken when S_d increases around the wire, and it would not produce the ordered capacitive family in Figure 3.

4.2 Why f_{res} falls with S_d

Thickening the jacket simultaneously (i) removes plasma dielectric near the wire (favoring an upward shift toward free space) and (ii) inserts series C_{sh} (favoring a downward phase-zero shift, as in PIP $f_{\phi 0}$ arguments). Under the $\text{Im}\{Z\}$ crossing metric on

this grid, series capacitance wins. On $\Delta x = 0.04$ m, $S_d = 2$ is already an 8 cm vacuum jacket—a strong capacitive load relative to a physical Debye sheath—which amplifies series- C domination.

4.3 Relation to Tu et al. and Song et al.

Tu et al. (2008) and Song et al. (2007) address high-voltage transmitting dipoles with self-consistent, time-varying sheaths. The present runs prescribe a static step jacket in fluid FDTD. We therefore do not claim Tu dynamic fidelity. The appropriate static bridge is $C_{\text{sh}}(r_s)$ (and later, in follow-on work, prescribed $\dot{r}_s$), not an upward Im-zero scoreboard.

4.4 Limitations

1. Free-space f_{res} on the same dipole remains an open upper anchor for the unloading narrative.
2. C_{eff} mixes sheath and plasma reactance; a pure gap extraction would require additional modeling (e.g., vacuum reference or circuit de-embedding).
3. Absolute coax comparison depends on r_{eff} ; the reported ratio ≈ 0.3 is for $r_{\text{eff}} = 0.23\Delta x$. Larger r_{eff} raises C_{sh} further; smaller r_{eff} improves absolute agreement but does not change the shared S_d trend.
4. Campaigns here are effectively unmagnetized; PIP upper-hybrid markers require $B \neq 0$.
5. Low-frequency C_{eff} exists only for $S_d = 2$ and 10. Intermediate widths enter through the dense band, where C_{eff} is a more mixed observable. That is a sampling choice, not evidence against those jackets.

5 Conclusions

A prescribed volumetric vacuum sheath in PF-FDTD loads a plasma-immersed dipole as a **series** capacitor because the jacket coats the wire and feed current must cross the gap by displacement current. CW measurements at all six thicknesses show monotonic capacitive loading with S_d . $\text{Im}\{Z\}$ zero-crossings decrease with S_d where they can be located, and band-averaged $C_{\text{eff}}(S_d)$ decreases in parallel with coaxial analytic $C_{\text{sh}}(S_d)$ at a scale factor ≈ 0.3 in the quasi-static window. These results validate series-gap load-

ing as the correct FDTD interpretation of a static jacket and provide the quantitative baseline needed before prescribing time-varying sheath boundaries or self-consistent charging.

Acknowledgments

The authors thank collaborators on the PF-FDTD sheath campaign for discussions of coupling diagnostics and impedance markers.

Open Research

Phasor tables supporting the figures are in `paper0_sheath_campaign/analysis/data/` (dense band: `cw_tu_summary.txt`; low-frequency: `cw_lowf_summary.txt`). Derived C_{eff} tables and figure scripts are in `paper1_static_sheath_capacitance/analysis/`. Solver source is the collab PF-FDTD repository (PEC-seeded `ApplySheath`). Replace this section with a persistent DOI before submission.

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